

On the Decidability of $\mathcal{ALCI}_{\text{RCC5}}$ (and $\mathcal{ALCI}_{\text{RCC8}}$) Concept Satisfiability: Split-Forest Models, a Conditional Decidability Theorem, and a Machine-Checked Reduction to One Keystone

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July 2026

Abstract

Whether concept satisfiability in $\mathcal{ALCI}_{\text{RCC5}}$ and $\mathcal{ALCI}_{\text{RCC8}}$ —the description logic $\mathcal{ALCI}$ with the RCC5/RCC8 base relations as roles under *abstract composition-table* semantics—is decidable has been open since Wessel’s 2002/2003 work. This paper reports a sustained, adversarially-reviewed, partly machine-checked attack on the problem, conducted by AI assistants under human direction. We do not settle the question. What we establish is: (i) a *conditional decidability theorem*—if the “live width” of models is bounded by a computable function of the concept (property F6), then satisfiability is decidable—whose soundness direction is *machine-checked in Lean 4*; (ii) that the entire remaining difficulty compresses into that single property F6, which we show is the *same fact* that blocks a proof of undecidability; (iii) a certified fragment of the local algebra (an RCC5 normal form) and an unconditionally decidable fragment (the $\forall$ PO-free concepts); and (iv) an honest map of the routes tried, the dead ends, and the missing hard pieces. We also report, candidly, on the methodology—adversarial “cold review,” machine verification, and a human-in-the-loop workflow—and on what frontier AI could and could not do.

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1 Introduction

1.1 AI disclaimer and authorship

The mathematical content of this paper was produced by AI assistants (Anthropic’s Claude models and OpenAI’s GPT-5.5), prompted, steered, and reviewed by the human author (Michael Wessel), who posed the problem, supplied the central intuitions and directions, orchestrated the review–repair cycles, and takes responsibility for the framing. **The results are unverified by human domain experts and should be read as AI-generated conjectures supported by machine checking and computational evidence—not as established theorems**, with the explicit exceptions of the parts stated to be kernel-checked in Lean 4 (the soundness pipeline, the abstraction’s faithfulness, and the RCC5 normal form), which are machine-verified but still human-unreviewed. We are deliberate about this labelling throughout: the standing one-line status is *strongly supported, not certified*. We regard the AI systems as having made contributions of a kind that, in a human collaborator, would warrant co-authorship, and we discuss this in §7.

1.2 Why the problem is hard, in one page

RCC5 relates two regions by exactly one of five relations: equal (EQ), proper part (PP), proper superpart (PPI), partial overlap (PO), or discreteness/disjointness (DR). Under *abstract composition-table* semantics a model is not a set of actual regions but a set with a total labelling of ordered pairs by these relations, required only to be converse-closed and *composition-closed*: if $x R y$ and $y S z$ then $x T z$ for some T the composition table allows.

The five relations split into two axes that behave completely differently.

- **The vertical axis** (PP, PPI: nesting) is *tame*. Composition can *force* vertical structure—the only four “determining” compositions (those returning a single relation) all run through PP or PPI—but an unbounded vertical chain folds into a finite loop, exactly as in ordinary modal logic. Infinite depth is not the problem.
- **The horizontal axis** (DR, PO: side-by-side) is *wild*. Composition can never *determine* a horizontal relation (PO is never a forced value), yet it can force two occurrences to be *distinct* (a shared horizontal anchor excludes EQ). So horizontal “crowds” of pairwise-distinct occurrences are the danger.

A finite decision procedure must record, at each interface of a model, the genuine branching it cannot reconstruct—the *live width*. Many forced edges are *shadows* (implied by composition from edges already present) and cost nothing; live width counts only the rest. The whole decidability question reduces to one property:

(F6) Is the live width of models bounded by a computable function of the input concept?

If yes, the certificate space is finite and satisfiability is decidable. And here is the knife’s edge: F6 is *the same fact* that blocks proving the problem *undecidable*. To force undecidability one would encode a rigid coordinate grid, which requires forcing a horizontal relation to a single value—exactly what the composition table cannot do (the “coincidence obstruction” Wessel identified in 2002/2003). Turn that looseness into a bound and one gets decidability; circumvent it and force a grid and one gets undecidability. The two open directions are one question seen from two sides, which is why neither has been settled in over twenty years.

1.3 Scope, contributions, and structure

This is both a technical report and a discussion piece. Its contributions are:

1. A *conditional decidability theorem* ($F6 \Rightarrow \text{decidable}$) whose soundness half is machine-checked in Lean 4 (§5.5), and the reduction of the whole problem to F6.
2. The identification of F6 as the shared keystone of the decidability and undecidability directions (§1.2, §8).
3. A certified RCC5 *normal form* and an unconditionally decidable $\forall\text{PO}$ -free fragment (§5.2, §5.6).
4. An honest, technique-organized account of every route tried, the counterexamples that killed the wrong ideas, and the lessons (§5), so that others need not repeat them.
5. A candid methodology report on adversarial AI-assisted mathematics (§7).

The paper is organized as follows. §2 states, in largely non-technical terms, where the problem stands after twenty-five years. §3 gives the history ($\mathcal{ALCT}_{\text{RCC}}$ from Cohn 1993 through Wessel 2002/2003 and Lutz–Wolter 2006) and carefully distinguishes our object from the neighbouring logics that *are* known undecidable. §4 explains why the standard undecidability reductions fail. §5—the core of the paper—surveys the approaches *by technique*: naive tableaux and blocking, finite-model and reduction attempts, the split-forest idea, the automata route, the Lean line culminating in F6, and the regular-cover pivot. §6 describes the working reasoners and a GIS example. §7 reports on the AI methodology. §8 concludes with what was learned, what would move the needle, an honest estimate of the chances of closing F6, and the project’s actual wins.

2 Current status: where we stand after twenty-five years

Decidability of $\mathcal{ALCT}_{\text{RCC5}}$ (and $\mathcal{ALCT}_{\text{RCC8}}$) concept satisfiability remains open. A current-literature check (July 2026) found no published resolution; the closest neighbours are decided in the other direction and are discussed in §3. What the present work adds is not an answer but a sharp reduction and a machine-checked surround.

The positive result—conditional decidability. *If the live width of models is bounded by a computable function of the concept (property F6), then concept satisfiability is decidable; and the soundness half of that reduction is machine-checked in Lean 4* (a valid finite certificate provably unfolds to a genuine RCC5 model of the concept, and the finite type abstraction is faithful). What is open is the antecedent F6 itself—equivalently the completeness direction, that every satisfiable concept has a *bounded* finite certificate.

What is certified (machine-checked, Lean 4, zero `sorry`):

- the *soundness* pipeline: a valid finite certificate unfolds to a genuine RCC5 model of the concept;
- the *faithfulness* of the finite (Hintikka) abstraction: satisfiability and abstraction-realizability coincide;
- the *RCC5 normal form* (forward): every strong-EQ RCC5 network is an ordered-disjoint structure—a strict partial order for PP with a downward-closed disjointness for DR, PO residual.

What is strongly supported but not certified: F6, the single open keystone—now sharpened to a precise combinatorial dichotomy (does some concept force unbounded “identity-selector minors”?), with the auxiliary uniformization property W2’ shown to *fold into* it.

Unconditional by-products (independent of F6):

- the $\forall$ PO-free fragment is *decidable*—concepts with no $\forall$ PO subformula, still retaining $\exists$ PO, $\forall$ DR, $\exists$ DR, and all vertical modalities, so a genuinely expressive spatial fragment;
- the certified RCC5 normal form above;
- working reasoners (a cover-tree tableau) that reproduce, 23 years later, the GIS taxonomy of Wessel’s original report;
- the standard undecidability routes shown blocked, with the coincidence obstruction identified as the shared fact.

The honest bottom line. After roughly thirty repair rounds and fifteen adversarial reviews the project has reached a genuine *local optimum*: the local algebra is exhausted (and in one case certified), the soundness side is machine-checked, and the whole remaining difficulty is one well-posed lemma. We offer the work not as a resolution but as a two-decade open problem *reduced, honestly and checkably, to its irreducible core*.

3 History of $\mathcal{ALCI}_{\text{RCC}}$ and why the known results do not settle it

4 Why standard undecidability reductions fail

5 Overview of approaches, by technique

We organize the attack by *technique*, in roughly the order tried, not by round or date. Each subsection follows the same shape: the high-level idea; the concrete techniques investigated; where and why they fail (with the examples that made the failure clear); pointers to the papers, probes, and reasoners; and the lessons.

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- 5.2 Finite-model and reduction attempts
- 5.3 The split-forest idea
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